DeepWaterSeg: High-Resolution Satellite Imagery Analysis for Water Body Extraction Using ResNet-U-Net

Sk. Khaja Mohiddin Basha
Assistant Professor(CSE)
Narasaraopeta Engineering College
Narasaraopeta, India
skbhasa@gmail.com

R. Anjaneya Kumar
Department of CSE
Narasaraopeta Engineering College
Narasaraopeta, India
anjaneyakumar1804@gmail.com

K. Durga Prasad
Department of CSE
Narasaraopeta Engineering College
Narasaraopeta, India
komaragiridurga319@gmail.com

S. Matheen Baba
Department of CSE
Narasaraopeta Engineering college
Narasaraopeta, India
Munnasyed769@gmail.com

S. Peer Mohammad Shaahid
Department of CSE
Narasaraopeta Engineering college
Narasaraopeta, India
Shaikshaheed409@gamil.com

***Abstract* -- Accurate identification of water bodies is an important aspect of ecological oversight, disaster management, and resource allocation. DeepWaterSeg is a deep learning technique for precise and accurate extraction of water bodies from high-resolution satellite images, including ponds, lakes, rivers, and more. It uses a ResNet50-based U-Net architecture, which enables precise mapping of water bodies.**

Keywords – Water Body Segmentation, Satellite Imagery, Water body mapping, Disaster Management, Resource allocation

I. Introduction

Water bodies like lakes, rivers, ponds, and fishponds are vital components of the freshwater ecosystem. Their presence supports biodiversity, regulates climate, and provides essential resources for agriculture, industry, and domestic use. Managing these water bodies is also a crucial part of disaster management, especially in recurrent flooding regions. Therefore, the identification, extraction, and mapping of these water bodies are really important which promotes efficient disaster management and resource allocation

From [1],[2] and [3], 71% of water bodies are saline water and out of remaining 29% freshwater we can make use of only 1% and remaining is drained to sea water bodies

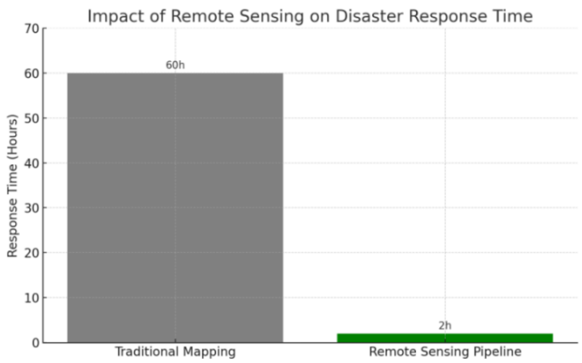
i. Fresh water statistics

Category	*Approximate % of Freshwater*
Inaccessible Freshwater	*~99.697%*
Accessible Freshwater	*~0.3%*
Human Use	*~0.00026%*
Wasted to Oceans	*~0.0026*

As per [4] Average disaster response time with tradition water body identification is 2-3 days which face a steep dip to 2hrs with remote sensing identification of water bodies.

Water body segmentation tasks usually suffer from multiple problems, including segregating water bodies from complex landscapes and terrains. One more noticeable problem is the colour of water. It is important to distinguish between varying water colours across different landscapes.

ii. Average disaster response time



To address these challenges, we propose a deep learning solution which make use of Resnet50 with U-Net architecture with some efficient preprocessing steps by delivering this project efficient water body extraction leads to loss less or minimal loss disaster management Resource management and fresh water ecosystem preservation

In the current research we have decided to opt restnet50 based U-Net architecture feed with 2800 high quality satellite imagery and trained over 30 epochs. Literature review, Dataset, Model architecture, Mathematical notations, Model performance and Conclusion and discussed later in the paper.

II. Literature Review

Accurate water body segmentation tasks used to produce minimal results with traditional methods. However, in the recent past, due to new approaches, these tasks have been experiencing drastic improvements over the last couple of years, primarily due to advancements in Deep Learning—especially techniques associated with imagery data.

Xu et al. [5] came up with DMLU-Net which is specially trained for water body segmentation. This method used **dual** encoders to improve accuracy and detect even thin edges. This approach enhances the water body extraction in **varying** resolution images.

Chagas et al. [6] used Sentinel imagery for mapping and identification of small water bodies with the help of temporal data and spectral indices. This is not entirely preprocessing but provides **valuable** insights that can be integrated with **neural** networks.

Ye et al. [7] introduced a high-precision water body reconstruction technique. This approach **combines** 3D **stereo images** with **semantic** segmentation, which enhances the accurate water body segmentation in high **vegetative** landscapes such as cities and urban areas.

Attya et al. [8] proposed a **deep learning** approach that integrates the convolutional **neural** networks with attention mechanism on satellite images. This model showed **robust** results with high-resolution images, precisely **distinguishing** between shadows, cloud reflections, etc.

Jonnala et al. [9] came up with AER U-Net, an attention-based U-Net architecture with **residual** blocks and skip **connections**. It enriches the **attention mechanism**. Later on, the model's robustness is tested against **Sentinel-2** imagery.

Hence, in this model, we have used ResNet50 as the backbone for the encoder and a U-Net model as the decoder, which improves the performance of water body segmentation.

III. Dataset

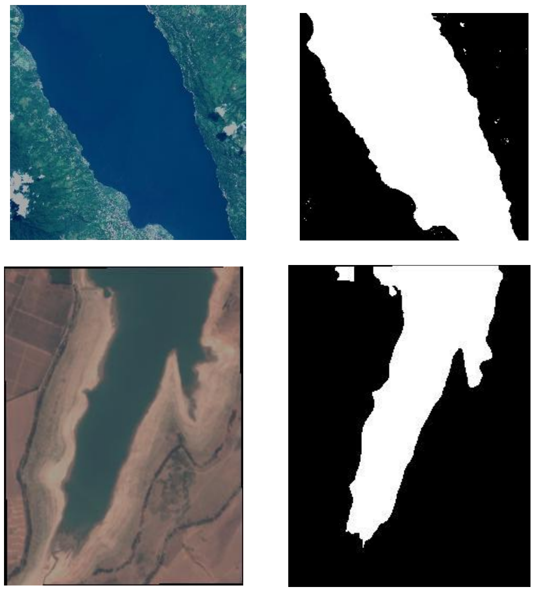
This model has trained on High resolution satellite imagery as using of low resolution images has multiple setbacks that includes improper identification of small and thin water bodies and imperfect boundary identification which are neither ideal nor tolerable. The data set available at https://www.kaggle.com/datasets/franciscoescobar/satellite-images-of-water-bodies consists of over 2800 high resolution images and corresponding ground truth masked images as well for supervised learning

High resolution imagery is perfect for identifying thin and narrow waterbodies such as curvy rivers, small ponds and lakes. Due to the use age of high resolution images, enable the identification of urban water body, irrigation land water bodies and wet land water bodies . High resolution imagery enables model to distinguish between water bodies and similar looking shadows which will eventually promotes model's accuracy

The data set not only includes large number of images it also does have lot of diverseness and variability of data. The data set contains of high resolution imagery that captures water bodies from different landscape and differently coloured water (blue colour fresh water, green colour algae and leachy water) which promotes the model to generalize more and can perform well on real life imagery as well which is both essential and commendable for practical deployment.

iii. Sample Images from dataset

Images Ground truth masks



IV. Data preprocessing

Data set consists of high resolution satellite imagery of different sizes as deep learning model expect it's inputs having fixed shape to run filter on the images , following preprocessing steps are performed before model training

- Converting images to RBG from BGR this meets the expectations of the RESNET50 mode as RESNET50 is pretrained on image data, we can use simple instruction with Computer Vision for RBG conversion
- Converting masks to grayscales in which masked image will be mapped to binary values which promotes the efficient water body extraction

Masked image Pixel values $\in \{0, 1\}$

- Resizing images and mask, as we are using RSENET50 with UNET architecture which expects it's input to have fixed size for applying filter we must make sure that all images are of same size and shape
- Normalizing pixel values, by dividing each pixel with 255 we can map our current image to new space where each pixel ranges from 0 to 1

Pixel values $\in [0, 1]$

- Expanding mask dimensions: Grayscale masks are converted to 3D by adding a channel dimension, making them compatible with the model's output shape (height, width, 1).

Here are few images before and after performing preprocessing tasks that includes all the techniques mentioned above

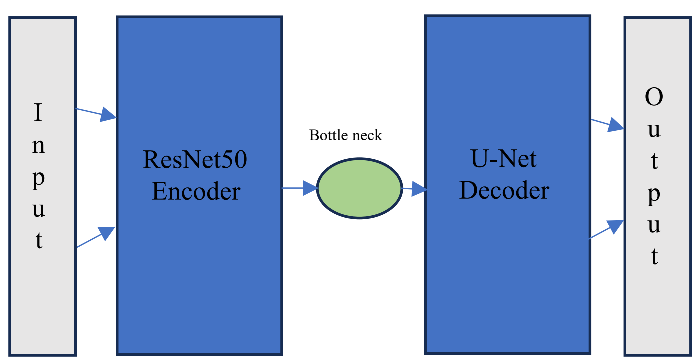
Before Preprocessing After Preprocessing

V. Satellite images

iv. Masked images

V. System Architecture Overview

vi. High level model Architecture



ResNet50 U-Net model is designed with a Encoder and Decoder. As Resnet50 is pretrained on Images it can perform feature extraction precisely hence, ResNet50 is used as an Encoder and U-Net like architecture is used to as a Decoder Here is the model high level architecture

ResNet50 encoder maps the input image to lower dimensional space while capturing complex details , A bottle neck is a layer between Encoder and Decoder and a U-Net like Decoder construct the image from lower dimensional space to higher dimensional space with minimal loss of details

Detailed architecture of ResNet50 -U-Net

Now let's dive in to more detailed low level design of ResNet50 U-Net Architecture

Input layer: Input layer accepts images of dimensions 256 X 256 X 3 where height=256 pixels , Width = 256 pixels and has 3 colour channels (RGB)

ResNet50 Encoder

ResNet50 encoder consists of 5 stages in the following stages

First stage of ResNet50 Encoder is a convolutional layer uses a input layer of dimensions of 256 X 256 X 3 with rectified linear unit activation function and make use of 7 X 7 convolutional layer with 64 filters with stride of 2 this reduces the spatial dimension to half of it's original dimensions and create a 128 X 128 X 64 dimensional output. This transforms raw RGB image into a feature map due to use of large kernel (7 X 7) allows the network to extract the board spectral patterns

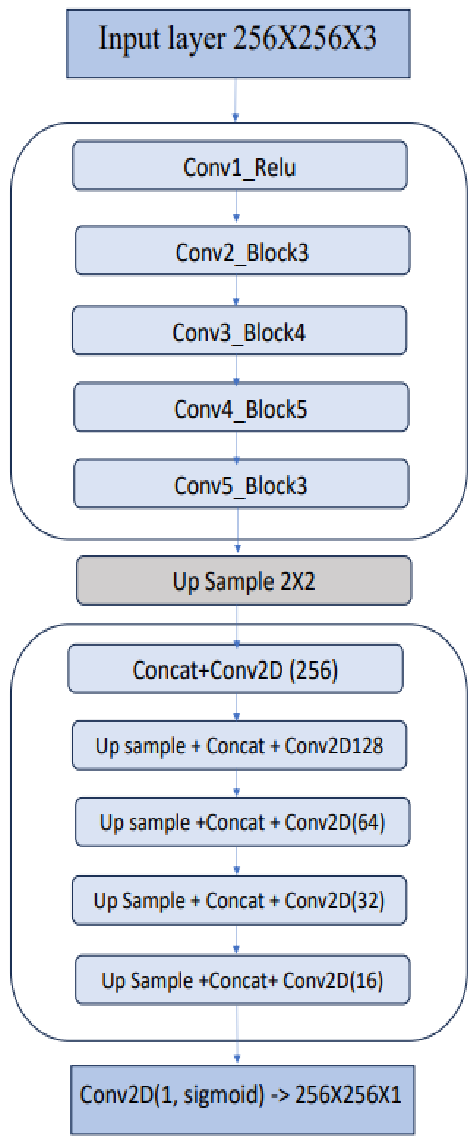
Second stage of ResNet50 Encoder: This stage contains **Conv2_block3_out** where the 64 feature map of Conv1_relu output is feed for second stage This stage contains 3 residual blocks each having two 3 X 3 convolutional layers with 64 filters followed by batch normalization and Relu activation and generates a output of dimension 64 X 64 X 256, The input is reduced to 64 pixel feature map

Third stage of ResNet50 Encoder: This stage contains **Conv3_block4_out** is feed with output of Convo2_block3_out (64 X 64 X 256) and has 4 Residual blocks each having 3 X 3 convolutional layers and followed by batch normalization and Relu activation which generates a 32 X 32 X 512 feature map the depth is increased to 512 filters which depicts a deeper feature representation.

Fourth stage of ResNet50 Encoder : This stage uses a **conv4_block6_out** that contains 6 residual blocks of each having 3X3 convolutional layers this stage is feed with 32 X 32 X 512 Feature map of previous stage output and this stage maps the input to 16 X 16 X 1024 which is more deeper feature extraction.

Fifth stage of ResNet50 Encoder: This is the final stage of encoder, **conv5_block3_out** contains 3 residual blocks of size 3 X 3 which maps the spatial dimension to 8 X 8 and filter size of 2048 which can represent the most deepest features from the image

vii. Model architecture diagram



Bottle Neck (Bridge)

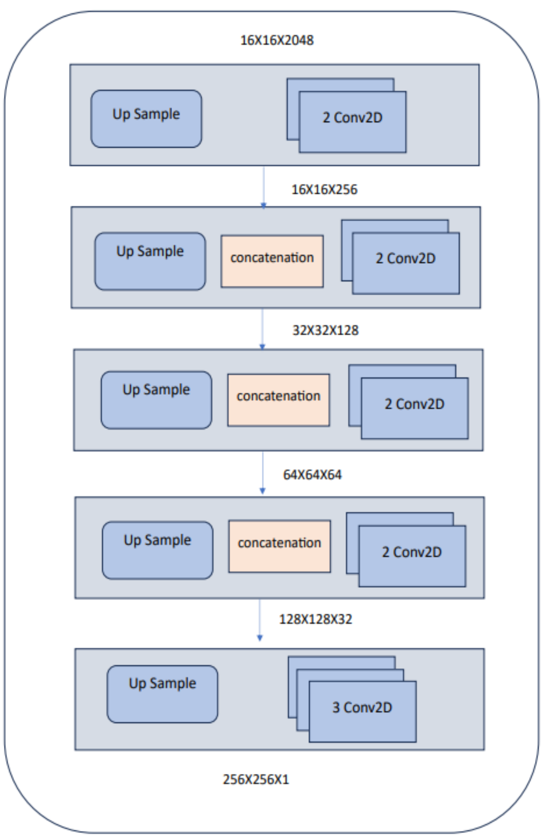
The Bottle next is the deepest layer in the model and serves as a critical connection between the encoder and decoder paths. It takes the final output of the encoder, with a spatial size of (8×8×2048) and enriches the spatial size to 16X16 with same sematic wealth. The layer extracts high-level semantic features with the largest receptive field, encoding the most abstract and informative features of the input. These detailed features are subsequently sent to the decoder for progressive reconstruction of the segmentation mask.

U Net Decoder

U-Net Decoder in this model consists of 6 stages where in each stage the input will be mapped into new higher dimensional space the output of Encoder dimension is 8 X 8 X 2048 is mapped to 256 X 256 X 1

Here is more detailed view of U-Net decoder that includes Up samples, Concatenation each stage of U-Net decoder

viii. Decoder Architecture



Decoder stage 1 (8×8 → 16×16):

Decoder begins with the encoder final output of dimension (8X8X2048) which has lavish sematic details (2048) but very low in Spatial data (8X8) hence in this layer the Spatial data is up sampled to 16x16 dimension and reduces the sematic detailing from 2048 to 256 consists of 3 layers including 1 concatenate and 2 Conv2D.

Decoder Stage 2 (16X16→ 32X32):

Second stage of decoder consists of 4 layers that includes one up sample one concatenate and two Conv2D layers. This layer maps the input to higher spatial information (32X32) from 16X16 and reduces sematic information as well from 256 to 128 frames

Decoder Stage 3(32X32→64X64):

Third stage of U-Net encoder consists of four layers (one up sample, one concatenate and two Conv2D) this stage maps the spatial information from 32X32 to 64X64 and reduces the sematic information from 128 frames to 64 frames

Decoder Stage 4(64X64→128X128):

Stage four of decoder enriches the spatial information from 64X64 to 128X128 and reduces the sematic information to 32 frames from 64 frames this stage consists of four layers same as satge3 and stage2 (one up sample, one concatenate and two conv2D layers)

Decoder Stage 5(128X128→256X256):

The last stage of U Net encoder – stage 5 consists of 4 layers unlike stage 4, 3 and 2 stage 5 consists of one up sample and 3 conv2D layers this layer maps the input spatial information to 256X256 from lower spatial information (128X128) and drastically reduces the sematic information from 32 to 1. The final dimension of the data is 256X256X1.

Model parameters

Parameter	Value
Input shape	(256X256X3)
Learning Rate	0.0001
Optimizer	Adam

Loss	Binary Cross Entropy
Epochs	30
Batch Size	8
Validation split	0.2
Activation	Relu and sigmoid at last layer
Filters	256 → 128 → 64 → 32 → 16
Smooth	0.000001

Mathematical Notations of ResNet50-U-Net Architecture:

Input:

N number of images of dimension (256X256X3) normalized to 0's and 1's

$$\boldsymbol{X} \in \boldsymbol{R}^{N\times 256\times 256\times 3} \quad \textbf{(normalized to [0, 1])}$$

Where, X is the input image that belongs to whole data set on dimension with 256X256 spatial information and 3 sematic information

Encoder Layers (ResNet50):

The current layer's feature map is the sum of the transformed previous layer's feature map and the current layer's original feature map.

$$F^l = H^l(F^{l-1}) + F^l$$

Where, F1 is the current Encoder stage, H1 is the current Encoder stage's convolution, batch normalization, and ReLU.

For 3ed stage it is something line

$$F_3 = H_3(F_1) + F_2$$

U-Net like Decoder:

$$U(F_l) = \text{UpSampling2D}(F_l)$$

$$C_u = \text{concatenate}(U(F_l), S_k)$$

$$F_{l+1} = \text{ReLU}(W * C_u + b)$$

In upscaling the input features are mapped to new dimension by increasing Spatial dimensions (Height and weight) and reducing Filters

Output:

$$\hat{Y} = \sigma_{\text{sigmoid}}(W_{\text{out}} * + b_{\text{out}}),$$

$$\text{where}, \ \sigma_{\text{sigmoid}}(x) = \frac{1}{1 + e^{-x}}$$

The final output layer is Conv2D uses as Sigmoid activation function where Wout is the output of the convolution kernel, F_{last} is the feature map produced after the final set of Convolutional layers and B_{out} is the output layer Convolutional operation

VI. Implementation

To implement this model python is used as language as Python most widely used programming language for development of AI powered applications and model and famous deep learning API, The TensorFlow Keras API is used in here due to it availability of pretrained models on imagery data , Google Colab is the ide used to train the model as it provides Nvidia GPU which can accelerate the model training process and attain results quickly.

The key idea is to use a pretrained ResNet50 model as it trained on ImageNet data which is helpful for the Water body Segmentation tasks.

This segmentation task is based on a U-Net architecture that uses a ResNet50-based encoder. Due to its strong feature extraction capabilities, ResNet50 was chosen as the encoder. By using the pretrained ResNet50 model on ImageNet, we leverage transfer learning to enhance performance and accelerate convergence.

The Resnet50 encoder has 5 major stages as following conv1_relu, conv2_block3_out, conv3_block4_out, conv4_block6_out, and conv5_block3_out where only first stage has no Residual blocks and all of the stages has Residual blocks which maps the image from higher spatial space to lower spatial space with higher filters which can help with detecting all minute details and more complex details

ReLU(x)=max(0,x)

A bottle neck layer is used as a bridge between the Encoder and decoder that ensures the abstracted features are up sampled back to higher dimension segmentation map

The U-Net based Decoder does map the lower dimension map to higher dimension segmentation mask. In our model: upsample1 with ReLU, concatenate with conv4_block6_out, Conv2D(256), upsample2, concatenate with conv3_block4_out, Conv2D(128), upsample3, concatenate with conv2_block3_out, Conv2D(64), upsample4, concatenate with conv1_relu, Conv2D(32), upsample5, Conv2D(16), and at the final layer Conv2D with 1×1 filter with sigmoid activation function is used.

Sigmoid activation function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

For validation metrics like Accuracy, Precision, Dice coefficient, Intersection over Union (IOU) and a loss coefficient are used these will be discussed more precisely in the results section

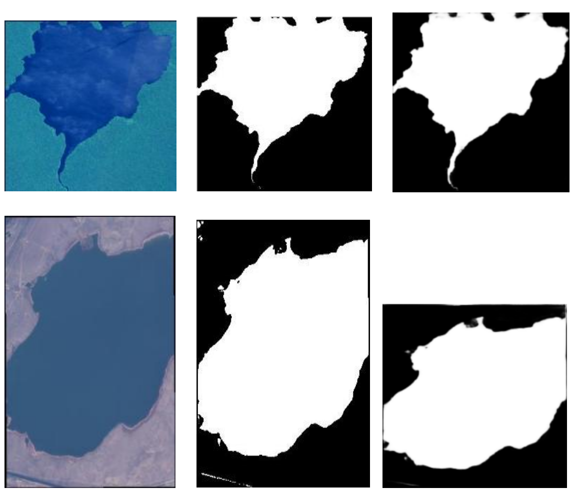
VII. Results

The Resnet50 with U- Net architecture model has made significant results and the ground truth masked images are pretty identical with the model predictions

Here are two examples how model predictions are identical with ground truth images

ix. Ground Truth v/s predicted mask

Image **Ground Truth Mask** **Predicted Mask**



After 30 epochs the model has made a significant progress in all metrices values but Precision metric value stand out with ground breaking result of 96% with a drastic achievement especially in water body segmentation tasks other metrics such as Dice coefficient , Accuracy, Recall and IOU got good results as well with 89%, 88.5%,83% and 80% respectively .

Precision: Precision is a quantitative measurement (evaluation metric) used to find the ratio between the True positives with total number of predicted positives , As this is a Segmentation task this precision value is often referred as pixel wise precision (True positive pixel v/s total predicted positive pixels),e with ResNet50 with U-Net Architecture 96% pixel wise precision is archived.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Accuracy: Accuracy is ratio between number of true predictions against total number of predictions. For Image segmentation task it is called as Pixel accuracy which means Number of truly predicted pixels per total number of predicted pixels. This model has archived 88.5% accuracy which is astonishing

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

Recall: Recall is the ratio of correctly predicted positive pixels (True Positives) to the total number of actual positive pixels in the ground truth. This segmentation model has archived Recall of 83% which noticeable and depicts model's good performance

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Dice Coefficient: Dice Coefficient is widely used metric in image segmentation tasks it is a balance between Recall and accuracy , Dice coefficient is harmonic mean of Accuracy and Recall. ResNet50 U-Net model has archived Dice coefficient of 89% which depicts the model's precise performance .

$$\text{Dice Coefficient} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}}$$

IOU: Intersection over union (IOU) is one another widely used image segmentation task metric, which depicts how much area does predicted mask covers the area in true mask. This model has archived 80% IOU results.

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$

Loss: In deep learning Loss is a metric that show how far does the predicted value is from the true values The Resnet50 with U-Net architecture model loss is limited to 13% only which is a noticeable take away in water body segmentation task Here lose is balance between Dice loss and Binary cross entropy loss which is a balanced metric and lambda(λ) is the balancing factor

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \cdot \log(\hat{y_i}) + (1 - y_i) \cdot \log(1 - \hat{y_i})]$$

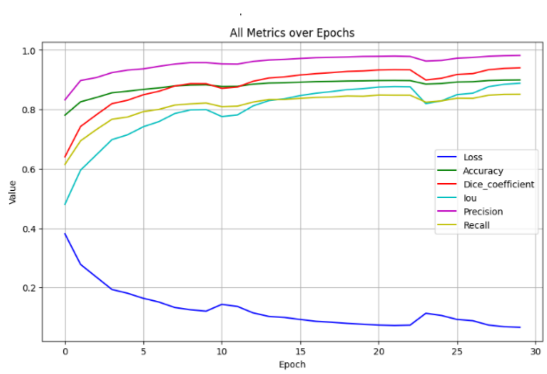
$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y_i} + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y_i} + \epsilon}$$

$$L_{Total} = (1 - \lambda) \cdot L_{BCE} + \lambda \cdot L_{Dice}$$

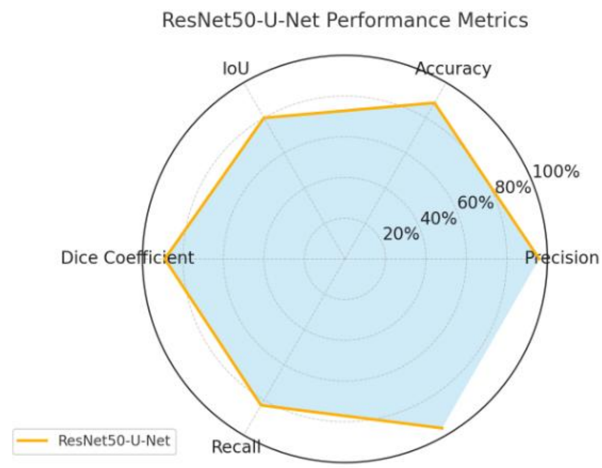
Here is a line plot of model performance against epochs where metrices includes Precision, Dice coefficient, Accuracy, Recall Intersection over union and Loss.

Here is the overall model performance with in a radar plot with metrics that includes Precision, Accuracy, Recall, Dice and IOU.

x. Evolution Metrics



xi. Model overall performance



VIII. Comparison with Existing System

Recent advancements in deep learning have paved the way for smoother image segmentation tasks. Recent works include DMLU-Net [5], which proposed a hybrid architecture with dual encoders that gained significant attention and improved performance in detecting thin edges—crucial for segmentation tasks in urban areas. However, it is considerably more complex. SD-Net [10] achieved competitive performance but at the cost of increased training time and architectural complexity.

Our ResNet50-U-Net model addressed these drawbacks and also attained good model performance with lightweight and simpler model architecture and attained 96% precision, 89% Dice coefficient, and 90% Intersection over Union (IoU)—outperforming models such as those in [5] and [8] which reported with 91% precision. This means better pixel prediction which makes up images close to truth ground values.

Unlike models in [6][7][13] that are trained on specific terrain, our model has been trained on a diverse dataset that includes complex terrains and different water bodies such as rivers, ponds, and lakes, etc. This enhances the real-life performance and generalizes very well, especially while working on small water bodies [6] and extracting water body and ground boundary extraction [12].

Unlike other models with complex architecture, our model is simpler as well as efficient and generalized very well, which is suitable for real-life usage.

IX.

X. Conclusion

We make use of Deep learning techniques and successfully trained a ResNet50-U-Net architecture Water body extraction using high resolution satellite images that includes different waterbodies such as rivers, Lakes and ponds with varying terrains and landscape.

This project indeed has lot of technical advancements and also excels average disaster response time, Sustainable development, Resource management, Fresh water preservation, Fresh water habitats and Fresh water ecosystem which can potentially contribute to conservation of fresh water life and promotes accountability.

The integration with deep learning for remote sensing task paves a way for automation solution for water body management and promotes sustainable development goals, efficient disaster management and environmental protection.

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